

Introduction to Comp Architecture

Quiz # 3

FA23-BIS-129

Dua Fatima.

Question #1:

The principal locality of reference states that during program execution, the CPU tends to access a small portion of memory repeatedly within a short period of time.

Temporal Locality

if memory location is accessed once it is likely to be accessed soon again.

e.g.
for ($i=0; i < 100; i++$)

sum = sum + i;

now sum is accessed many times repeatedly

Spatial Locality

if one memory location is accessed nearby memory locations are likely to be accessed soon.

e.g. accessing elements of an array sequentially.

$A[0], A[1], A[2]$

as elements are stored in consecutive memory locations.

Question #2:

1) memory in bytes addressable:

$$64 \text{ KB} = 64 \times 1024 = 65536$$

$$\text{number of address bits} : 2^{16} = 65536$$

$$\text{Bits equal for memory address} = \boxed{16 \text{ bits}}$$

2.2 ^(step 1) Each line has 8 bytes
 $8 = 2^3$

3 word bits.

Step 2: cache has 32 lines
 $32 = 2^5$

5 line bits

Step 3: Total address bits = 16
 tag bits = $16 - (5 + 3)$
 $= 16 - 8$
 $= 8 \text{ tag bits}$

2.3 Bit Division

	Tag	Line	Word
Bit range	15-8	7-3	2-0
Bit division	8 bits	5 bits	3 bits

Question # 3 :

Policy	Advantage	Disadvantage.
Write Through	main memory is always updated, making memory consistent and reliable.	shows slower performance occur every write updates both cache and main memory.
Write Back	Faster performance since data is written to main memory only when the block is replaced.	More complex and risk of data loss if power failure occurs before memory update.

- write through is chosen when simplicity and data reliability are important
- write back is chosen when high performance and reduced memory traffic are required

Computer organization & Architecture.

Assignment #3

PA23-BCS-129

Isha Fatima.

Question #1:

a) Direct Mapping

In each block of main memory can be placed in only one specific cache line

- Placement Rule: $(\text{Main Memory Block}) \bmod (\text{No. of Cache Lines})$
- Hardware Complexity: Low complexity.
- Conflict miss tendency: High.

b) Fully Associative Mapping:

Any memory map can be replaced by any cache line

→ Placement rule:

- block may occupy any available line in cache.

→ Hardware complexity:

- Very high complexity and expensive in hardware due to parallel comparisons

→ Conflict miss tendency: Very low

c) Self Associative Mapping:

In this the cache is divided into sets, each the memory block maps to specific set but can be placed in any line within that set.

Placement Rule: $\text{Set Number} = (\text{Memory Block}) \bmod (\text{No. of Sets})$

Hardware complexity: moderate complexity.

Conflict miss tendency: moderate / lower than direct mapping but higher than fully associative

Question #2:

2a) Word offset bits:
block size = 4

$4 = 2^2$
→ 2 bits are needed for offset bits.

b) Set Index Bits
no of set = 8

$8 = 2^3$
3 bits at set index.

c) Tag bits
total address bits = 16 bits

$$\begin{aligned}\text{tag bits} &= 16 - 3 - 2 \\ &= 16 - 5 \\ &= 11\end{aligned}$$

11 bits therefore

d) Address field Division : 11 bits | 3 bits | 2 bits

final Representation : Tag (15-5) | Set (4-2) | Offset (1-0)

Question #3:

Victim cache:

It is a small associative mapping placed between the main cache and main memory. It temporarily stores blocks that are removed (evicted) from the main cache.

organized? → It is organized as usually used in direct mapped cache and consists of a fully associative entries and stores recently replaced cache blocks.

performance improvement: victim cache helps reduce conflict misses in direct mapped cache.

e.g. two frequently used memory blocks may continuously replace each other in direct mapped cache. instead of going to main memory, the replaced block may still exist in victim cache, taken access.